

# **Coodinated multiple sluicing in Korean**

An experimental approach

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## Multiple sluicing in Korean

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## Multiple sluicing

- Sluicing is the elliptical construction where a *wh*-remnant remain after deletion process, which has a corresponding counterpart in the context, a correlate (Ross 1969, Lasnik 2001, Merchant 2001).
  - (1) He is writing something, but you can't imagine [what]. (Ross 1969: 252)
- Sluicing also permits to appear with multiple remnants, a structure referred to as **multiple sluicing** (Takahashi 1994).
  - (2) John gave something to someone, but I want to know [who] and [what]. (Hoyt & Teodorescu 2012: 86)
- Multiple sluicing occurs in languages with *wh*-in-situ questions, similar to Japanese, and it is also found in *wh*-fronting languages, such as German or Slovenian (Merchant 2001).

## Coordinated multiple sluicing in Korean

- In this study, we focus on multiple sluicing in a coordinate structure (henceforth, CMS) in Korean.
  - (3) a. Someone saw something, but I can't remember, [who] \*(or) [what]. [English]  
(Lasnik 2014: 8)
  - b. nwukwunka-ka mwuenka-lul kacyewa-ss-nuntey,  
Someone-NOM something-ACC bring-PST-but,  
'Someone brought something,  
[nwu-ka] (**kuliko**) [mwues]-i-nci molu-keyss-e.  
who-NOM (and) what-COP-QUE not.know-PRES-DECL  
but I don't know who (and) what.' [Korean]
- In Korean, the two *wh*-remnants appear to be coordinated, and the copula verb *i-* is obligatorily followed by the interrogative-clause marker with the latter remnant.
- Distinct from English, Korean allows an optional coordinating conjunction.

## Research questions

- The goal of this research is
  - to offer a broader implications by exploring potential sources of ellipsis.
  - to experimentally test interpretations and answerhood conditions.

## **Similarities and Differences between English and Korean CMS**

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## Multiple wh-remnants

- Both Korean and English allow wh-remnants, but English is more limited, being more acceptable when they are coordinated or include a wh-adjunct:

- (4) a. John gave something to someone, but I want to know [who] and [what]. (Hoyt & Teodorescu 2012: 86)
- b. \*I know that everybody bought something. I just don't know [who] [what].
- c. I know John was talking with somebody about something, but I don't know [who with] [about what]. (Richards 1997: 164)
- (5) chinkwu-tul-i phathi-eyse mwues-ul calmos mek-ko paythal-i na-ss-nuntey,  
friend-PL-NOM party-LOC what-ACC wrong eat-CONJ stomachache-NOM get-PAST-QUOT  
Some employees experienced stomach issues after the team dinner,
- a. [nwu-ka] (kuliko) [mwues]-i-nci molu-keyss-e  
who-NOM (and) what-COP-QUE not.know-FUT-DECL  
I don't know who and what.
- b. [mwues-ul] (kuliko) [elmana]-i-nci molu-keyss-e  
what-ACC (and) how.much-COP-QUE not.know-FUT-DECL  
I don't know what and how much.

## Coordination of unlike categories i

- Both in English and Korean, remnants could be either an argument or an adjunct, allowing coordination of unlike categories, irrelevant of their linear orders.

- (6) a. Someone ate something, but I don't remember **who** and **what**. (*argument-argument*)  
 b. Someone ate the pie, but I don't know **who** and **when**. (*argument-adjunct*)  
 c. Someone ate the pie, but I don't recall **when** and **who**. (*adjunct-argument*)  
 d. The pie was eaten, but I don't know **by whom** and **when**. (*adjunct-adjunct*)
- (7) chinkwu-tul-i phathi-eyse mwues-ul calmos mek-ko paythal-i na-ss-nuntey,  
 friend-PL-NOM party-LOC what-ACC wrong eat-CONJ stomachache-NOM get-PAST-QUOT  
 Some employees experienced stomach issues after the team dinner,
- a. [nwu-ka] (kuliko) [mwues]-i-nci molu-keyss-e.  
 who-NOM (and) what-COP-QUE not.know-FUT-DECL  
 I don't know who and what. (*argument-argument*)
- b. [nwu-ka] (kuliko) [elmana]-i-nci molu-keyss-e.  
 what-NOM (and) how.much-COP-QUE not.know-FUT-DECL  
 I don't know who and how much. (*argument-adjunct*)

## Coordination of unlike categories ii

- c. [elmana]        (kuliko) [nwuku]-i-nci molu-keyss-e.  
how.much-COP (and) who-QUE not.know-FUT-DECL  
I don't know what and how much. *(adjunct-argument)*
- d. [way] (kuliko) [elmana]-i-nci molu-keyss-e.  
why (and) how.much-COP-QUE not.know-FUT-DECL  
I don't know why and how much. *(adjunct-adjunct)*

## Coordinating conjunction in multiple *wh*-remnants

- In English, the presence of the coordinating conjunction ‘and’ is obligatory in order for multiple sluicing to be grammatical (Citko & Gračanin-Yuksek 2020).

- (8) a. \*John sang something, but I don’t remember **what where**.  
 b. John sang something, but I don’t remember **what and where**.

(Adapted from Citko & Gračanin-Yuksek 2020: (14a))

- In Korean multiple sluicing, however, the coordinating conjunction *kuliko* seems to be optional.

(9)

kil-ey mwulken-ul ttelettuly-ess-nuntey,  
 street-LOC belonging-ACC drop-PST-CONJ,  
 ‘I dropped something at the street,

- a. **encey kuliko eti**-eyse-i-nci molu-keyss-e.  
 when and where-LOC-COP-QUE not.know-FUT-DECL  
 but I don’t know when and where.’  
 b. **encey eti**-eyse-i-nci molu-keyss-e.  
 when where-LOC-COP-QUE not.know-FUT-DECL  
 but I don’t know when and where.’

## Two possible readings: English

- In English, only the Single-Pair (SP) reading is available in coordinated multiple *wh*-sluicing (Whitman 2002, Citko 2013, Citko & Gračanin-Yuksek 2020).

(10) A: Someone left this house, but I don't know **who and when**.

B: Mimi left in the afternoon.

*(Single-Pair reading)*

B': \*Mimi left in the afternoon and Lee left at midnight.

*(Pair-List reading)*

(Adapted from Citko & Gračanin-Yuksek 2020: 10)

## Two possible readings: Korean

- In Korean, however, both Single-Pair and Pair-List (PL) readings are available in multiple sluicing (Jung 2018).

(11) A: mwuenka-ka pwunmyeng pang-eyse epse-cin-kes kath-a.  
 something-NOM definitely room-LOC disappear-become-KES seem-DECL  
 It seems that something has disappeared from the room.

B: **mwe-ka** kuliko **encey?**  
 what-NOM and when  
 ‘What and why?’

B: yelsoy-ka cinancwu-pwuthe an-poy-e.  
 key-NOM last.week-since not-see-DECL  
 ‘A key has not been found since last week.’ (SP Reading)

B’: yelsoy-ka cinancwu-pwuthe anpoy-ess-ko kapang-un ecey-pwuthe anpoy-ess-e.  
 key-NOM last.week-since not.see-PST-CONJ bag-TOP yesterday-from not.see-PST-DECL  
 ‘A key has not been found since last week, and my bag since yesterday.’ (PL Reading)

- This implies that the coordination of unlike wh-interrogatives implies both single and multiple events (Whitman 2004: among others)

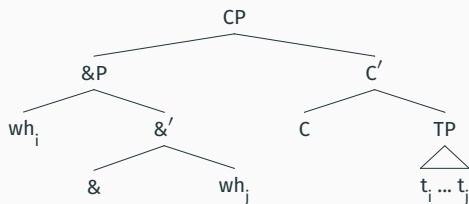
## Previous analyses

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## Mono-clausal structure

- A mono-clausal (multidominant) structure allows a single node (e.g., V, T, C) to be multiply dominated and have two mothers by a Parallel Merge operation (Citko 2005, Gračanin-Yuksek 2007, Citko 2013, Citko & Gračanin-Yuksek 2020).

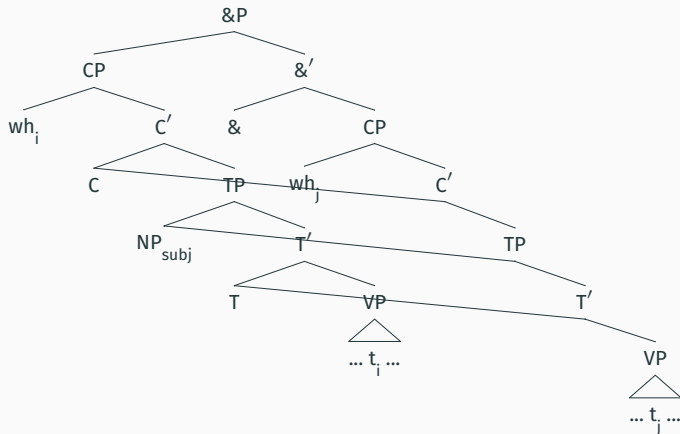
(12)



## Bi-clausal structure i

- The bi-clausal structure coordinates the remnants at the CP level, sharing the lexical material in the antecedent as a whole, except for the wh-phrases (Gračanin Yuksek 2007, Citko 2013).
- Bi-clausal non-bulk sharing structure (Gračanin Yuksek 2007)

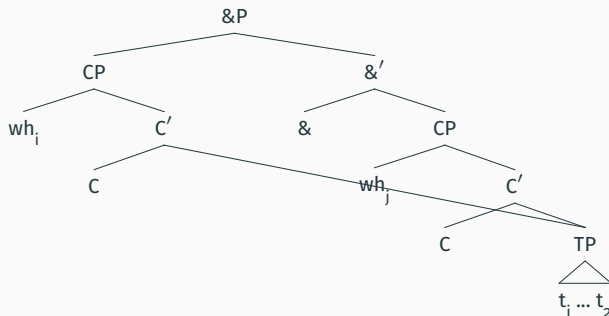
(13)



## Bi-clausal structure ii

- Bi-clausal bulk sharing structure

(14)

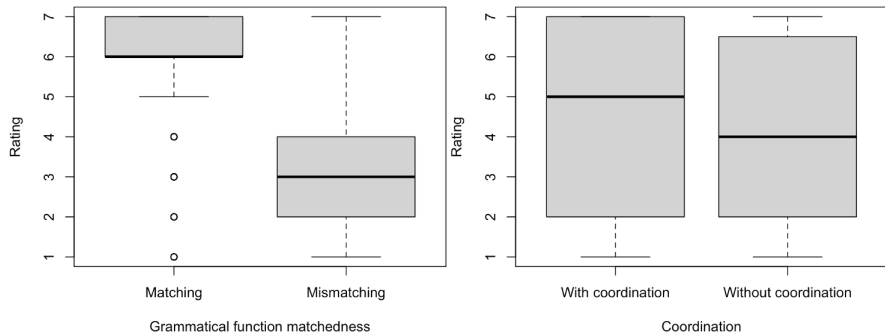


- If these strategies are applied to the coordinated multiple sluiced wh-remnants, it raises doubts whether they can account for some syntactic or semantic restrictions in Korean.

## Experiments: Pilot study

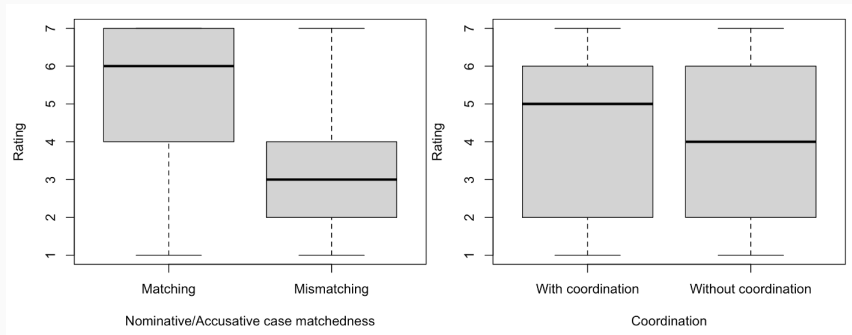
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## Experiment 1: Grammatical function matchedness



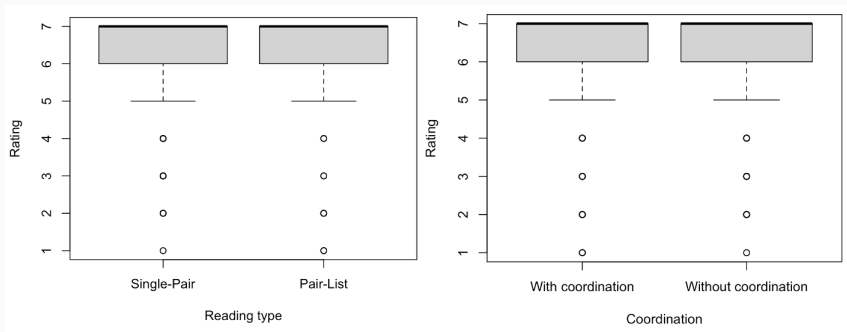
- Cumulative link mixed model analysis with parsimonious model shows that the effect of grammatical matching ( $\beta = 5.515$ ,  $SE = 0.467$ ,  $z\text{-value} = 11.800$ ,  $p\text{-value} = < 2e-16$ ) is significant while that of coordination is not ( $\beta = 0.185$ ,  $SE = 0.316$ ,  $z\text{-value} = 0.586$ ,  $p\text{-value} = 0.558$ ).
- Their interaction was not significant ( $p\text{-value} = 0.505$ ), either.

## Experiment 2: Case marking matchedness



- The effect of case marker matching ( $\beta = 3.170$ ,  $SE = 0.345$ ,  $z\text{-value} = 9.180$ ,  $p\text{-value} = < 2e-16$ ) is significant, and that of coordination is significant as well ( $\beta = 0.662$ ,  $SE = 0.259$ ,  $z\text{-value} = 2.554$ ,  $p\text{-value} = 0.011$ ).
- However, their interaction was not significant ( $p\text{-value} = 0.1439$ ).

## Experiment 3: Reading preferences



- Both of the effect of reading preference ( $\beta = -0.172$ ,  $SE = 0.283$ ,  $z\text{-value} = -0.610$ ,  $p\text{-value} = 0.542$ ) and that of coordination is not significant ( $\beta = 0.035$ ,  $SE = 0.284$ ,  $z\text{-value} = 0.123$ ,  $p\text{-value} = 0.902$ ).
- Their interaction was not significant ( $p\text{-value} = 0.094$ ), either.

## Mean analysis in reading preference

|                  | Total rating means | Coordination      |              |
|------------------|--------------------|-------------------|--------------|
|                  |                    | Coordination Type | Rating means |
| Single-Pair      | 6.2604             | <i>kuliko</i>     | 6.262        |
|                  |                    | <i>No kuliko</i>  | 6.260        |
| Pair-List        | 6.2598             | <i>kuliko</i>     | 6.261        |
|                  |                    | <i>No kuliko</i>  | 6.259        |
| <i>kuliko</i>    | 6.2616             |                   |              |
| <i>No kuliko</i> | 6.2613             |                   |              |

- A paired t-test shows that the mean ratings between Single-Pair (mean = 6.2604) and Pair-List (mean = 6.2598) is not significantly different ( $t = -0.6077$ ,  $df = 407$ ,  $p\text{-value} = 0.272$ ).

## Further experiments in consideration

- We need to observe the following considerations in future experiments:
  - optional transitive verbs: *mekta* ‘eat’, *ilkta* ‘read’, *pota* ‘see/watch’, *ssuta* ‘write/use’
  - ARG-ADJ matching between a remnant and its correlate
  - sprouted remnant cases
  - more sophisticated experiments settings for SP-PL reading tasks
  - optional *kuliko*

## Summary

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## Summary

- Multiple sluicing in Korean shows not only similarities but also differences from English in that both allows argument or/and adjunct *wh*-remnant pairs, regardless of the overtness of correlate in the antecedent site.
- However, they differ in the optionality of the coordinating conjunction, especially when the argument remnant restricts the sluicing with certain case markers, which also allows for variability in the two readings.
- As in previous analyses, syntactically distinct structures result in different interpretations that multiple sluicing can have.
- Unlike English, where multiple sluicing is limited to the Single-Pair reading, preferences in the two readings determined by acceptability ratings can account for not only syntactic structure but also semantic behaviors of Korean multiple sluicing.

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